

# Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau

Phase 3 – Project Design Phase

## 1. Problem – Solution Fit

Each identified problem (from the business scenarios) is mapped to the Tableau feature that addresses it:

| Problem                                                          | Scenario                                             | Tableau Solution                                                                   |
|------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------|
| Delayed detection of declining interest in products/ingredients  | Scenario 1: Monitoring Consumer Preferences          | Real-time filterable dashboard with trend visualizations (line charts, bar charts) |
| Slow response to negative reviews or safety issues               | Scenario 2: Addressing Product Concerns              | Drill-down views by product/brand with alerts on rating/review patterns            |
| Inability to anticipate emerging trends and innovate proactively | Scenario 3: Predictive Analysis & Product Innovation | Historical trend analysis, forecasting visuals, and story-driven insights          |

## 2. Proposed Solution

Cosmetic Insights is a Tableau-powered analytics dashboard and story that consolidates brand, pricing, and skin-type data into a single interactive interface. Stakeholders can filter by brand, category, skin type, and price range to monitor trends in real time, investigate product concerns as they emerge, and use historical patterns to guide future product and marketing decisions. The dashboard and accompanying story are published to Tableau and embedded into a lightweight Flask web application for easy stakeholder access.

## 3. Solution Architecture

The solution follows a linear data-to-decision architecture:

- 1 Data Source Layer: Cosmetics dataset (Excel/CSV) containing brand, category, price, rank, and skin type fields.
- 2 Data Preparation Layer: Data cleaning, pivoting, and calculated fields created within Tableau (e.g., Price Tier, Rank Category).
- 3 Analytics Layer: Tableau Desktop/Public used to build unique visualizations — bar charts, box plots, heat maps, scatter plots, treemaps.
- 4 Presentation Layer: Interactive dashboard (with filters) and a multi-scene story built around the three business scenarios.

- 5 Delivery Layer: Dashboard and story published to Tableau, then embedded via Tableau's embed/JS API into a Flask web application.
- 6 User Layer: Brand managers, analysts, and regulatory stakeholders access insights through the web app.

This completes Phase 3 (Project Design), covering: Problem – Solution Fit, Proposed Solution, and Solution Architecture.